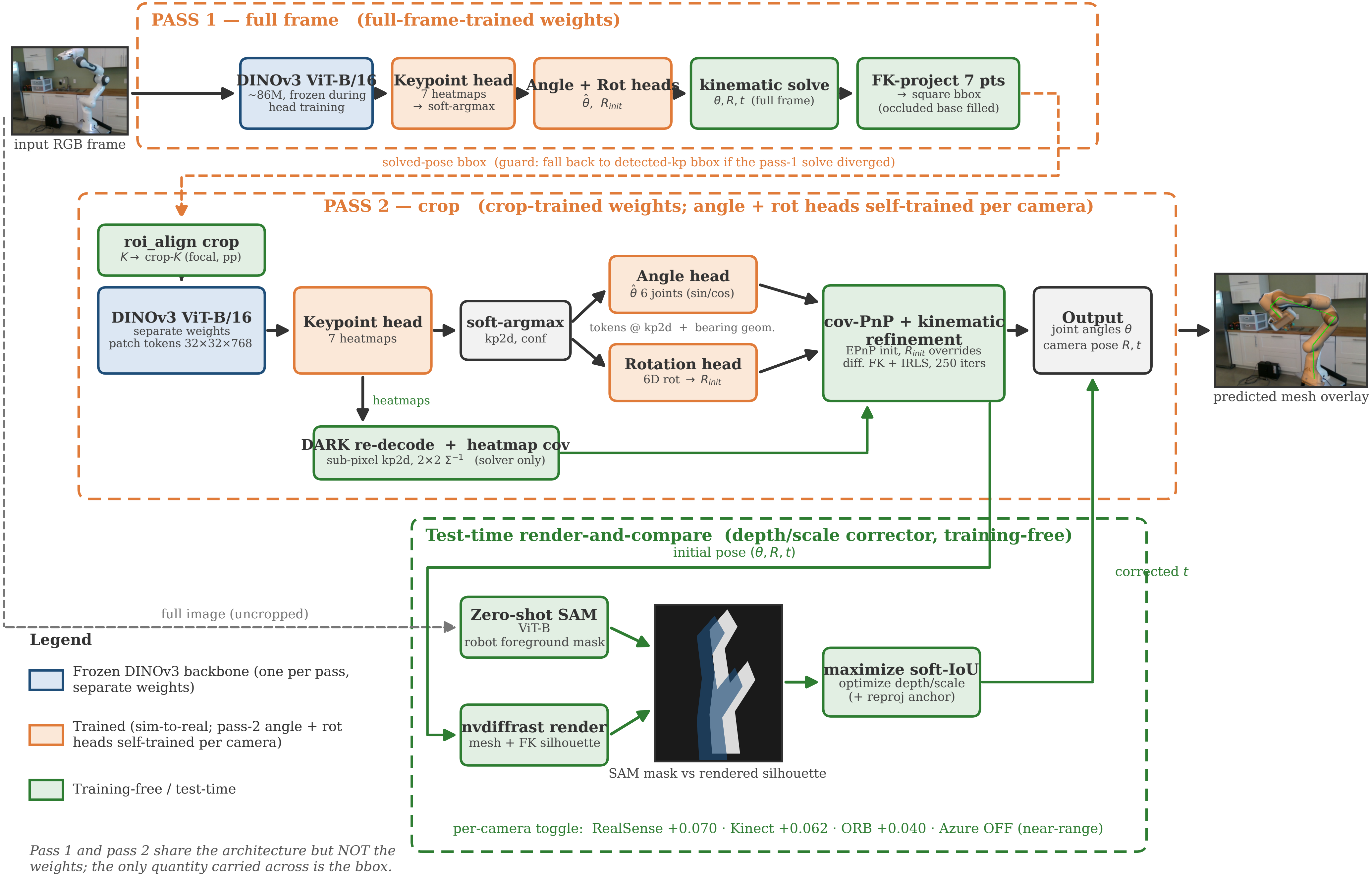
PASS 1 — full frame (full-frame-trained weights)
input RGB frame
DINOv3 ViT-B/16
~86M, frozen during head training
Keypoint head
7 heatmaps
→ soft-argmax
Angle + Rot heads
$\hat{\theta}$, R_{init}
kinematic solve
θ, R, t (full frame)
FK-project 7 pts
→ square bbox
(occluded base filled)
solved-pose bbox (guard: fall back to detected-kp bbox if the pass-1 solve diverged)
PASS 2 — crop (crop-trained weights; angle + rot heads self-trained per camera)
roi_align crop
$K \rightarrow$ crop-K (focal, pp)
DINOv3 ViT-B/16
separate weights
patch tokens 32×32×768
Keypoint head
7 heatmaps
soft-argmax
kp2d, conf
Angle head
$\hat{\theta}$ 6 joints (sin/cos)
tokens @ kp2d + bearing geom.
Rotation head
6D rot → R_{init}
cov-PnP + kinematic refinement
EPnP init, R_{init} overrides
diff. FK + IRLS, 250 iters
Output
joint angles θ
camera pose R, t
predicted mesh overlay
heatmaps
DARK re-decode + heatmap cov
sub-pixel kp2d, 2×2 Σ^{-1} (solver only)
Test-time render-and-compare (depth/scale corrector, training-free)
initial pose (θ, R, t)
corrected t
full image (uncropped)
Zero-shot SAM
ViT-B
robot foreground mask
nvdiffrast render
mesh + FK silhouette
SAM mask vs rendered silhouette
maximize soft-IoU
optimize depth/scale
(+ reproj anchor)
per-camera toggle: RealSense +0.070 · Kinect +0.062 · ORB +0.040 · Azure OFF (near-range)
Legend
Frozen DINOv3 backbone (one per pass, separate weights)
Trained (sim-to-real; pass-2 angle + rot heads self-trained per camera)
Training-free / test-time
Pass 1 and pass 2 share the architecture but NOT the weights; the only quantity carried across is the bbox.